Murtaza S. Ahmed

Undergrad Engineer | Distributed Systems, ML Systems & Infra (Backend focus)

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Focusing on AI infrastructure, particularly GPU cluster orchestration, inference systems, model routing, resource allocation and workload scheduling.

Professional Experience

Undergraduate Research Assistant (Edge AI Systems) — **Dec 2025 -- Present**

Department of Mechanical Engineering (MJCET) — *Hyderabad, TG*

- Selected for an institution-funded R&D program under faculty supervision.
- Built a complete edge AI system on NVIDIA Jetson Nano 4GB Developer Kit, integrating an IMX719 camera and soil sensors for real-time field operation.
- Developed the on-device inference pipeline and executed real-time inference under compute and memory constraints.
- Achieved 40–70 ms latency (14–20 FPS) using TensorRT (FP16) optimization.
- Processed camera and sensor data continuously at 5–10 Hz with low-latency performance.
- Designed a stable execution loop enabling reliable operation over 4–6 hour field deployments.

Undergraduate Researcher (R&D Program) — **March 2026 -- Present**

Department of Mechanical Engineering (MJCET) — *Hyderabad, TG*

- Design and implementation of an Adaptive Multi-Objective Policy Engine for Resilient Orchestration in Heterogeneous LLM Clusters.
- Researching adaptive workload scheduling and hardware-aware resource allocation strategies for heterogeneous LLM inference systems.
- Exploring fault-tolerant orchestration approaches for distributed AI workloads across mixed compute environments.

Technical Contributor (AI & ML) — **April 2025 -- July 2025**

Horizon Research Group — *Remote*

- Contributed technical research regarding AI/ML workloads and inference performance metrics.
- Investigated distributed scheduling strategies and model routing logic for scale-out neural architectures.

Builder & Systems Architect — **July 2025 -- Present**

Verity (Decentralized Verification Platform) — *Hyderabad, TG*

- Designing and implementing a system to verify digital media provenance using blockchain and decentralized storage.
- Integrating AI modules for manipulation detection and automated claim extraction.
- Developing media provenance infrastructure using distributed storage systems including IPFS and Arweave.
- Building concurrent ingestion pipelines and immutable audit logging mechanisms for ML-based verification workflows.

Backend Infrastructure Engineer — **Sept 2024 -- Present**

Freelance Consulting — *Hyderabad, TG*

- Migrated client applications to containerized Linux deployment environments with automated CI/CD workflows.
- Designed backend APIs and optimized PostgreSQL schemas for improved concurrency and deployment reliability.

Projects

Smart Agri Four-Legged Bot (SAFL-B) (*C, Python, UART/Serial, I2C, GPIO, Hardware Control*) — **Dec 2025**

- Awarded 1st Place at the MAKEFORHYDERABAD 2-Day Makeathon for autonomous robotics systems integration.
- Implemented deterministic UART/Serial communication protocols for multi-joint motor control.
- Developed telemetry stabilization routines to reduce terrain-induced sensor and image noise.

Deterministic Workspace Provisioning (*Bash, Linux, Arch Linux, Podman, Git*) — **June 2025**

- Built automated provisioning pipelines for lightweight Linux workstation deployment using Arch Linux and shell automation.
- Configured container runtimes (Docker/Podman) and developer tools to deploy reproducible workspace layers.

Technical Skills

Languages:	C, Python, Bash, PostgreSQL, Git
Systems:	Linux, Arch Linux, Docker, Kubernetes, Podman
Hardware & Interfaces:	UART/Serial, I2C, GPIO

Education

Bachelor of Engineering in Mechanical Engineering **May 2028**
Muffakham Jah College of Engineering & Technology (Osmania University) *Hyderabad, TG*

- Relevant Coursework: Scientific Computing, Embedded Systems Programming, System Modeling & Simulation